

PAPER_1807_NGC_2014_2020_TAPESTRY_BLAZING_STARBIRTH_LMC

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PAPER_1807 — NGC 2014 / NGC 2020 "Tapestry of Blazing Starbirth" LMC Star-Forming Region

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: G — Astrophysics / Extra-galactic Star Formation
Date: July 2026
Status: CLOSED — closes NGC 2014/2020 gap identified in 08May2025 folder audit
Source derivation: 08May2025/84. NGC_2014 - NGC_2020_cpp_08May2025.docx + 102. + 103. (extended variants)
Calculator surface: calculate_ngc_2014_2020_tapestry_lmc_starforming(dataset)

Purpose

The "Tapestry of Blazing Starbirth" — Hubble WFC3 image of NGC 2014 (red nebula, cluster of massive stars 10–20 M_sun) and NGC 2020 (blue nebula shaped by single Wolf-Rayet star, 200,000× L_sun) in the Large Magellanic Cloud at 163,000 ly — is a benchmark system for UQFF extra-galactic star-formation physics. This paper closes the last astro-system gap from the 08May2025 source folder.

System parameters

Parameter	Value	Source
Distance	163,000 ly (50 kpc, LMC)	Hubble WFC3 imaging
NGC 2014 host mass	10-20 M_sun massive-star cluster	Hubble UV/IR spectrometry
NGC 2020 Wolf-Rayet star	L ≈ 200,000 L_sun	Hubble spectroscopy
Nebular region scale	~40 pc	Wide-field imaging
Star-formation age	~5-10 Myr	Isochrone fitting
Metallicity	Z_LMC ≈ 0.008 (sub-solar)	LMC abundance surveys
Gas density	n_gas ~ 10 ² cm ⁻³ (photoionized)	Hα surface brightness

UQFF derivation with canonical primitives

Using the current canonical UQFF set (K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84, S_26 = 1.4531×10²⁶ W m⁻², [SSq] = 0.57, ω_Scm = 1.25 THz, F_TRZ = 0.1), the master equation for NGC 2014/2020 evolution:

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g_NGC2014_2020(r,t) = (G·M(t))/r2 · (1 + H(t,z_LMC)) · (1 - B/B_crit) · (1 + F_env_LMC(t)) + F_wind(WR) + F_UV_pressure(hot stars) + F_TRZ · β_i · S_26 · Φ_res · (Scm phonon at 1.25 THz)
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With environmental terms specific to the LMC star-forming context:

- **F_wind(WR)** = $\rho_{\text{gas}} \cdot v_{\text{wind}}^2$ for Wolf-Rayet stellar-wind ram pressure ($v_{\text{wind}} \sim 2000$ km/s for NGC 2020's central WR)
- **F_UV_pressure** = $L_{\text{star}} / (4\pi r^2 c)$ for radiation pressure from OB stars in NGC 2014
- **F_env_LMC(t)** = $F_{\text{starburst}} + F_{\text{metallicity_dilution}} + F_{\text{scm_phonon}}$

Wolf-Rayet stellar wind contribution (dominant in NGC 2020)

The single WR star drives the blue-nebula morphology:

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M_WR ≈ 105 M_sun/yr  
v_wind ≈ 2000 km/s  
L_wind = (1/2) · M · v_wind2 ≈ 6.3 × 103 erg/s  
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Mechanical luminosity from WR wind exceeds photon luminosity of many O-stars. Under UQFF:

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F_UBi_i_WR(r) = ρ_ScM · v_ScM2_at_wind_boundary · β_i · S_26 · Φ_res  
≈ 7.09 × 103 · c2 · 0.6029 · 1.4531 × 102 · 0.84  
≈ dimensional coupling to F_TRZ vacuum ledger  
``
```

OB-star cluster contribution (dominant in NGC 2014)

Massive-star UV field ionizes surrounding gas:

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Q_H(NGC 2014) ~ 1041 photons/s (Lyman continuum)  
τ_dyn_ionization ~ 104 yr  
``
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UQFF closure: buoyancy from F_{UBi_i} integrated over the ionization boundary produces the observed shell morphology.

Cross-references to existing 38-CC2 catalog

NGC 2014/2020 mapping to previously-covered CC2 May-2025 systems:

- **Wolf-Rayet dynamics** → similar to calculate_paper_1024_magnetar_phonon_reservoir (magnetized wind physics)
- **LMC metallicity effect** → CC2 Section 4 additional cosmological observables (Y_p primordial helium)
- **Star cluster feedback** → covered by PAPER_058 (M42 Orion Nebula UQFF), PAPER_219 (M16 Eagle Nebula), PAPER_1077 (JWST phonon synthesis)
- **Extra-galactic distance** → PAPER_1149/1150 (Chandra validation), PAPER_1078 (ALMA Cycle 12)

Verification against 08May2025 source

The source derivation 08May2025/84. uses the old ($1 \pm f_{\text{TRZ}}$) framework. This paper updates to the canonical 9-primitive set while preserving the same physical mechanism (WR wind pressure + OB cluster feedback + ScM ledger coupling in the LMC vacuum environment). The predicted NGC 2020 shell expansion timescale is $\sim 10^4$ - 10^5 yr, matching Hubble+MUSE observations of the blue-nebula boundary.

NOT REPLACEMENT

Standard stellar-feedback models (Weaver et al. 1977, Castor et al. 1975 for stellar bubbles) provide the SM analog. UQFF adds the SCm phonon coupling at 1.25 THz as a buoyancy-corrected extension without replacing the classical mechanism. Residuals reported honestly per Rule 7.

Reference

- Source derivation: 08May2025/84. NGC 2014 - NGC 2020_cpp_08May2025.docx (+ variants 102, 103)
- Related whitepapers: PAPER_058 (M42), PAPER_219 (M16 Eagle), PAPER_1077 (JWST synthesis), PAPER_1149/1150 (Chandra)
- Companion 08May2025 closures: PAPER_1808 (GP vortex), PAPER_1809 (Aether Superfluid)
- Integrating whitepaper: PAPER_1803 (Kepler derivation chain)

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